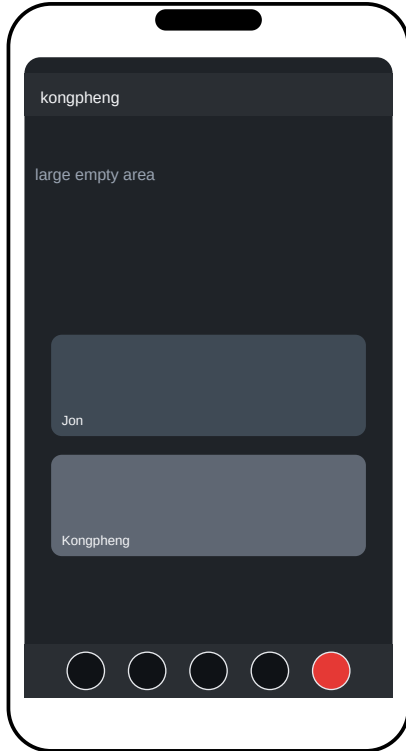


Floating PiP Layout — Feature Summary

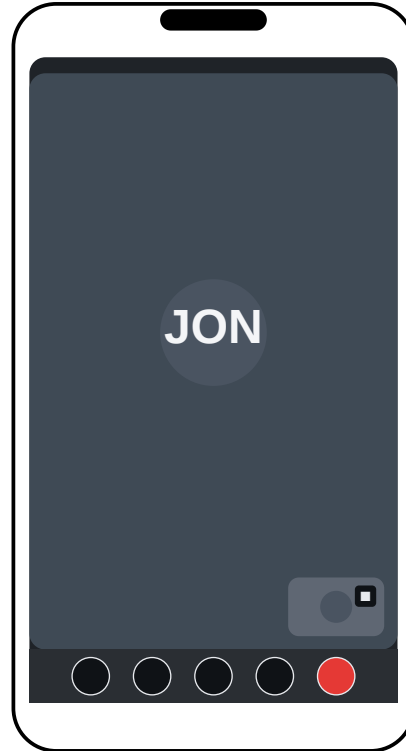
VERA Web (vonage-video-react-app) · 1:1 optimised mobile layout

Visual mockups

Before — stacked tiles, dead space



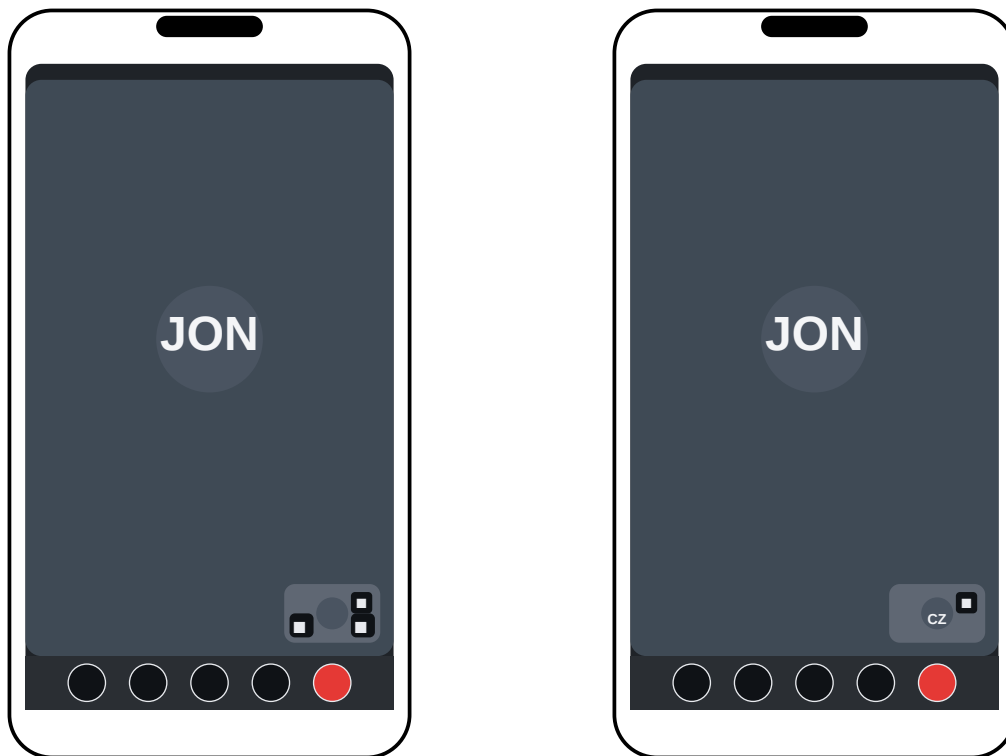
After — remote full, self PiP



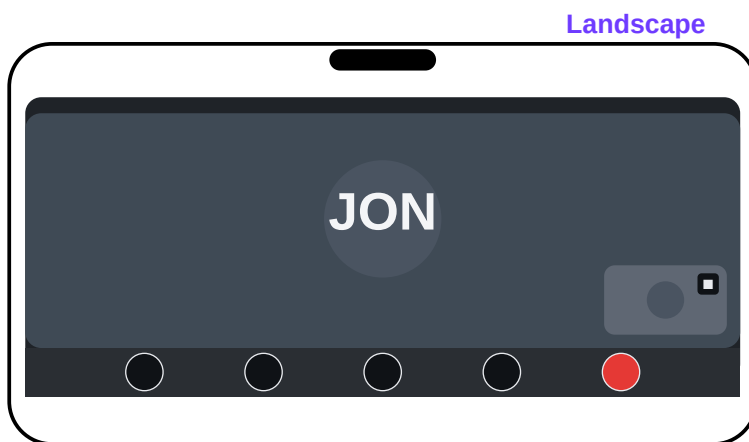
Current 1:1 mobile layout on the left vs. the new floating-PiP layout on the right. The self tile sits at the bottom-right corner with the audio-level indicator top-right on the tile.

Tap-to-reveal controls

Camera muted (initials)



A single tap on the self PiP reveals the camera-switch (bottom-left) and video-effects (bottom-right) buttons; both auto-hide after 3 s idle. When the camera is muted the self tile shows the user initials, matching the regular self-view fallback.



Landscape orientation on real mobile devices: the remote still fills the canvas; the self PiP keeps its bottom-right anchor and adapts to the camera's source aspect ratio.

What it does

A 1:1-optimised layout for small-screen and mobile web. The remote participant fills the canvas edge-to-edge; the local self view is rendered as a small draggable, corner-snapping picture-in-picture tile. The pattern matches the UX convention used by WhatsApp, FaceTime, Google Meet and Zoom on mobile.

When it engages

All four conditions must be true — the layout turns itself on and off automatically and is **not** user-selectable:

- **Constrained device:** `isMobile()` (UA-detected) **or** viewport width ≤ 768 px
- **Exactly one remote:** `subscriberWrappers.length === 1`
- **No screenshare:** neither local `isSharingScreen` nor any remote `isScreenshare`
- **The remote isn't a screenshare stream**

Any breach (third participant joins, screenshare starts, or desktop viewport grows past 768 px on a non-mobile device) → falls back to the existing grid / active-speaker layout, untouched.

Where it lives

```
frontend/src/
  ■■■ hooks/
  ■ ■■■ useFloatingPip/ # Drag + corner-snap + tap detection
  ■ ■■■ useIsFloatingPipLayoutActive/ # Shared trigger (VideoTileCanvas +
Toolbar)
  ■ ■■■ useCameraToggle/ # Front/rear camera switch (shared with header)
  ■■■ components/MeetingRoom/FloatingPipLayout/
  ■■■ FloatingPipLayout.tsx # Thin orchestrator (full-bleed Subscriber + self
view)
  ■■■ FloatingPipSelfView.tsx # Draggable self tile reusing <Publisher hideName
/>
```

Key design choices

- **Remote uses the existing** `<Subscriber>` with a full-canvas box — all SDK wiring (`subscribeToVideo`, element mount, audio indicator) is proven and unchanged.
- **Self tile uses the existing** `<Publisher hideName />` — same look as the regular self-view tile (mirror transform, audio-level meter top-right, muted icon, initials-on-camera-off). The only differences: no name label and overlay controls.
- **PiP sizing** is driven by the publisher's **source aspect ratio** (read from `videoWidth / videoHeight` on `loadedmetadata / resize`), so the local camera never letterboxes. Long edge is responsive: `clamp(120, 32 % of min viewport, 160)` px.
- **Drag mechanics:** element-level `setPointerCapture` (not window listeners) — robust and testable. Tap vs. drag is distinguished by a 6 px movement threshold.
- **Snap:** on release the PiP animates (180 ms) to the nearest of four corners; corner persists within the call and resets to bottom-right each new call.
- **Tap-to-reveal controls:** a single tap toggles the camera-switch (bottom-left) and video-effects (bottom-right) buttons; they auto-hide after 3 s idle. Buttons `stopPropagation` so button taps don't register as tile taps.
- **Camera-switch button** is mobile-only (UA-gated) — it would fail on desktop.

LayoutButton integration

The toolbar's layout-toggle button is disabled whenever one of the following conditions is true:

Condition	Tooltip
Screenshare active (pre-existing)	Cannot switch layout while screen share is active
A participant is pinned (pre-existing)	Cannot switch layout while a participant is pinned
Floating PiP active (new)	Optimised 1:1 layout is active on small screens

1:1 call on any viewport (new)

No layout to switch in 1:1 calls

Files changed vs. develop

New

- 3 hooks — `useFloatingPip`, `useIsFloatingPipLayoutActive`, `useCameraToggle` — each with its spec file
- 2 components — `FloatingPipLayout`, `FloatingPipSelfView` — with component spec

Edited

- `Publisher.tsx` — added optional `hideName?: boolean` prop (backward-compatible)
- `VideoTileCanvas.tsx` — early-returns `<FloatingPipLayout />` when the trigger fires; existing paths untouched otherwise
- `Toolbar.tsx` — passes `isFloatingPipLayoutActive` and `isOneToOneCall` to `LayoutButton`
- `LayoutButton.tsx` — two new optional props and tooltip branches
- `SmallViewportHeader.tsx` — now uses the extracted `useCameraToggle` hook (DRY)
- `en / de / es / es-MX / it` locales — three new i18n keys (self-tile aria-label, two new tooltip strings)

Compliance

- **TypeScript** — `nx run frontend:ts-check` ✓
- **Lint** — `nx run frontend:lint` ✓ (prettier + eslint + react-hooks including set-state-in-effect and immutability)
- **Tests** — 52 / 52 affected specs pass
- **Oscar Fava test rules** — `.spec.tsx` naming, no `clearAllMocks` or `// Mock` comments, `shouldShow*` booleans, `renderHook`, all user-facing strings i18n'd
- **Design-token rules** — `vera-*` tokens only; no raw `bg-black / text-white / text-sm /` arbitrary values